

NAVER LABS System Re-implementation for the IWSLT 2026 Instruction-Following Task

Anonymous ACL submission

Abstract

We re-implement the NAVER LABS IWSLT 2025 instruction-following pipeline (Lee et al., 2025) for the IWSLT 2026 Shared Task (constrained condition, short audio track), adapting it to the mandated components: SeamlessM4T-v2-large (Barrault et al., 2023) as the speech encoder and Qwen3-4B-Instruct (Team, 2025) as the LLM backbone. The three-stage approach—projector alignment, text-only LoRA pre-training, and multimodal merging—is preserved from the original design. We additionally construct 100k synthetic instruction-following examples across ten speech-centric task types (10k per task) from the provided corpora, suitable for further Stage 3 fine-tuning. Our primary model achieves COMET 0.781 on EN-ZH speech translation and BERTScore-F1 0.346 on English SQA on the MCIF benchmark. Code, training scripts, and generated data will be released on our GitHub.

1 Introduction

Multimodal speech LLMs such as SALMONN (Tang et al., 2024), Qwen-Audio (Chu et al., 2023), SpeechGPT (Zhang et al., 2023), and WavLLM (Hu et al., 2024) couple a frozen speech encoder with an instruction-tuned LLM (Ouyang et al., 2022) via a lightweight connector, enabling flexible multi-task inference through natural language prompts. The IWSLT 2026 Instruction-Following Shared Task (Organizers, 2026) formalizes this paradigm with the MCIF benchmark, evaluating unified models on ASR, multilingual ST (EN→{DE,IT,ZH}), and SQA. The NAVER LABS 2025 system (Lee et al., 2025) demonstrated a competitive three-stage pipeline in the IWSLT 2025 constrained setting (Abdulmumin et al., 2025), but was not publicly released.

We provide the first open-source re-implementation, adapted to the IWSLT 2026 constraints (SeamlessM4T-v2-large encoder,

Dataset	Task	Lang.	Stage(s)
CoVoST 2	ASR, ST	EN→{DE,IT,ZH}	1, 2*, 3
EuroParlST	ASR, ST	EN→{DE,IT,ZH}	1, 2*, 3
LibriSQA	SQA	EN (+ DE,IT,ZH [‡])	1 [†] , 2*, 3
NUTSHELL	S2TSum	EN	3
YTSeg	AChap	EN	3

Table 1: Training corpora per stage. *Text-only. [†]A.2 variant only. [‡]Machine-translated.

Qwen3-4B-Instruct LLM—replacing the LLaMA-3.1-8B (Dubey et al., 2024) backbone used in 2025). We further construct 100k synthetic instruction-following examples across ten speech-centric task types (Section 2.1) and ablate LoRA rank and learning rate configurations for Stage 2 text pre-training.

2 Task and Data

Shared Task. We participate in the **constrained condition**, short audio track (Organizers, 2026). Evaluation on MCIF uses WER (↓) for ASR, COMET (Rei et al., 2020) (↑) for ST, and BERTScore-F1 (Zhang et al., 2020) (↑) for SQA. Task instructions follow the natural-language prompt format of Lee et al. (2025).

Training Corpora. Core speech data is from CoVoST 2 (Wang et al., 2021) and EuroParlST (Iranzo-Sánchez et al., 2020) (ASR/ST) and LibriSQA (Huang et al., 2024) (SQA); multilingual SQA pairs in DE, IT, ZH are obtained by machine-translating LibriSQA via SeamlessM4T-v2. Stage 3 additionally draws on NUTSHELL (Maiekezu et al., 2024) for speech summarization and YTSeg (Retkowski et al., 2024) for audio chapter detection. Table 1 summarizes corpora per stage.

2.1 Synthetic Instruction-Following Data

We construct 100k synthetic examples (10k per task) from the provided corpora using open-weight

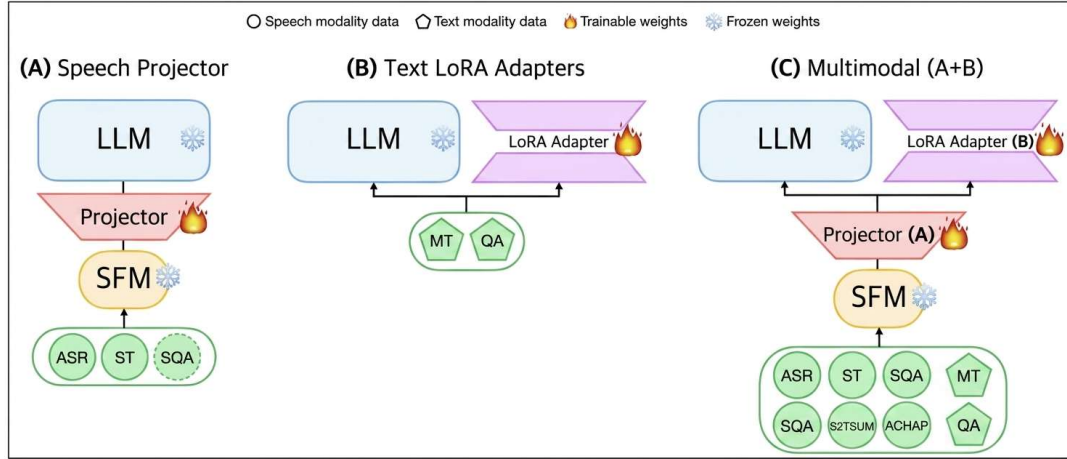


Figure 1: Three-stage training pipeline. Frozen modules: dashed border. Trainable: solid. Stage 3 jointly fine-tunes both projector and LoRA adapters.

Gemma models (Gemma Team, 2025). Seven text-grounded tasks are generated by Gemma-4-31B from reference transcripts: keyword extraction (T1), named entity recognition (T2), gist summarization (T3), topic labeling (T4), numeric QA (T5), and gist summarization in DE and ZH (T6–T7). Three audio-grounded tasks (T8–T10) are generated by Gemma-4-E4B-it directly from audio: vocal style description in EN, DE, and ZH.

For example, a NER target (T2) for the transcript “The Luks family eventually moved to Pottsville, in southern Pennsylvania” yields $PER=[Luks]$, $LOC=[Pottsville, S. Pennsylvania]$. A vocal style target (T8) yields “The speaker has a measured, confident tone, speaking at a moderate pace with a clear articulation.” This data will be released with our code.

3 System

Architecture. Our model follows Lee et al. (2025) and is illustrated in Figure 1. A frozen SeamlessM4T-v2-large encoder (Barrault et al., 2023) produces 1024-dim frame representations. A trainable projector downsamples by $3\times$ via frame averaging, passes through a 4-layer Transformer encoder (Vaswani et al., 2017), and projects to the LLM hidden size. A LoRA-adapted (Hu et al., 2022) Qwen3-4B-Instruct (Team, 2025) generates the response, with speech embeddings prepended at a $\langle | \text{speech} | \rangle$ placeholder.

Stage 1 — Projector Alignment. Encoder and LLM are frozen; only the projector is trained (4 epochs, $lr 1 \times 10^{-4}$, constant, AdamW (Loshchilov and Hutter, 2019)). **A.1** (ASR/ST): sampling 40%

ASR, 18% ST-DE, 24% ST-ZH, 18% ST-IT (CoVoST 2 + EuroParlST). **A.2** (ASR/ST/SQA): sampling 40% ASR, 10.5% ST-DE, 14% ST-ZH, 10.5% ST-IT, 25% SQA-EN (adds LibriSQA).

Stage 2 — Text-Only LoRA. No audio; projector frozen; LLM adapted via LoRA for 1 epoch. Sampling: MT 60% (20% each DE/IT/ZH, from CoVoST 2 and EuroParlST transcripts) and QA 40% (10% each EN/DE/IT/ZH, LibriSQA + machine-translated). Three configurations:

- **V1:** rank 8, $\alpha=16$, $lr 3 \times 10^{-4}$, attn+FF layers
- **V2:** rank 16, $\alpha=32$, $lr 1 \times 10^{-5}$, cosine, all-linear
- **V3:** rank 32, $\alpha=64$, $lr 2 \times 10^{-4}$, cosine, all-linear

Stage 3 — Multimodal Merge. Both the A.1 projector and V1 LoRA adapters are fine-tuned jointly; the speech encoder remains frozen. Sampling: 20% ASR, 10% each ST- $\{DE, IT, ZH\}$, 10% SQA-EN, 5% each SQA- $\{DE, IT, ZH\}$, 10% S2TSUM, 15% AChap. Each speech batch (ST, SQA) is immediately followed by a paired text-only batch (MT, QA) to prevent catastrophic forgetting. Projector lr : 1×10^{-5} (constant); LoRA lr : 3×10^{-4} (cosine); 2 epochs on $4 \times H100$.

4 Experiments

Setup. Stages 1–2 train on one H200 GPU; Stage 3 on $4 \times H100$ 80GB (DDP). Audio longer than 15 seconds is excluded due to memory constraints. Main evaluation uses the official mcif_eval tool; Stage 2 is separately evaluated on a 1k CoVoST 2 text subset (not MCIF-comparable).

Stage 1 & 3 Results (MCIF). Table 2(a) shows MCIF results. Stage 1 improves ST and SQA over

(a) MCIF Benchmark Results (constrained, short audio)								
Model	ASR	ST COMET ($\uparrow$)			SQA BERTScore-F1 ($\uparrow$)			
	WER ($\downarrow$)	DE	IT	ZH	EN	DE	IT	ZH
SeamlessM4T-v2-large (base)	21.49	0.674	0.723	0.638	0.153	0.151	0.158	0.125
A.1 Projector (ASR/ST)	28.90	0.698	0.727	0.763	0.186	0.185	0.186	0.186
A.2 Projector (ASR/ST/SQA)	37.94	0.661	0.709	0.732	0.267	0.296	0.251	0.289
Stage 3 (A.1 + V1 LoRA)	23.49	0.707	0.749	0.781	0.346	0.266	0.265	0.189

(b) Stage-2 Text Eval (1k CoVoST-2, not MCIF-comparable)					
Model	MT (DE)		MT (ZH)		SQA
	BLEU	COMET	BLEU	COMET	F1
Qwen3-4B (base)	25.48	0.824	2.03	0.865	0.482
+Stage 2 LoRA (V1)	31.69	0.852	13.18	0.880	0.652

(c) Stage-2 LoRA Rank Ablation					
Config	MT (DE)		MT (ZH)		SQA
	BLEU	COMET	BLEU	COMET	F1
V1 ($r=8, \alpha=16$)	31.69	0.852	13.18	0.880	0.652
V2 ($r=16, \alpha=32$)	30.94	0.851	13.97	0.880	0.650
V3 ($r=32, \alpha=64$)	32.58	0.853	15.34	0.880	0.600

Table 2: Experimental results. (a) MCIF benchmark; Stage 3 is our primary system. (b) Stage-2 text eval on 1k CoVoST 2; not MCIF-comparable. (c) LoRA rank ablation (lr: V1=3e-4, V2=1e-5, V3=2e-4); V1 used in Stage 3.

the SeamlessM4T-v2-large baseline at the cost of higher WER—consistent with Lee et al. (2025). A.2 boosts English SQA (0.267 vs. 0.186) but further degrades ASR. Stage 3 recovers ASR (23.49 WER), achieves the best ST COMET across all pairs (EN–ZH: 0.781), and strongly improves English SQA (0.346). Cross-lingual SQA remains lower due to sparse multilingual supervision.

Stage 2: Text Evaluation & LoRA Ablation. Since Stage 2 is text-only (no audio), its results are evaluated on the 1k CoVoST 2 text subset and are not comparable to MCIF. Table 2(b) shows the gain from LoRA (V1) over the base Qwen3-4B (Papineni et al., 2002; Post, 2018); Table 2(c) ablates LoRA rank. V3 (rank 32) achieves the highest MT scores; V1 (rank 8) yields the best SQA F1 and is selected for Stage 3.

5 Conclusion

We re-implemented the NAVER LABS three-stage instruction-following pipeline (Lee et al., 2025) for the IWSLT 2026 constrained setting, adapting it to SeamlessM4T-v2-large and Qwen3-4B-Instruct. Our Stage 3 model achieves COMET 0.781 on EN–ZH ST and BERTScore-F1 0.346 on English SQA on MCIF, with consistent improvements over projector-only baselines. The 100k synthetic dataset released alongside our code pro-

vides a natural extension point for richer Stage 3 fine-tuning or future reinforcement learning (Shao et al., 2024) with task-specific rewards and LLM-as-judge criteria (Kim et al., 2024).

Limitations

Audio longer than 15 seconds is excluded due to GPU memory constraints, potentially limiting performance on longer utterances. Cross-lingual SQA relies on machine-translated QA pairs, which may introduce noise. Stage 2 ablation numbers are evaluated on a text-only subset and are not directly comparable to MCIF results.

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A Synthetic Dataset: Task Definitions and Examples

Each of the 100k synthetic examples (10k per task) is paired with a natural-language instruction prompt; the model is trained to produce the target output. Tasks T1–T7 are *text-grounded*: Gemma-4-31B generates targets from reference transcripts. Tasks T8–T10 are *audio-grounded*: Gemma-4-E4B-it generates targets directly from speech audio. One example per task is shown below.

T1 — Keyword Extraction. *Definition:* Extract the most salient keywords from the spoken utterance as a comma-separated list.

Input: “These data components in turn serve as the ‘building blocks’ of data exchanges.”

Output: data components, building blocks, data exchanges

T2 — Named Entity Recognition. *Definition:* Identify and classify named entities (PER, ORG, LOC) present in the transcript.

Input: “Saunders was born in Farnborough, Hampshire, the son of a British Army officer.”

Output: PER:[Saunders], ORG:[British Army], LOC:[Farnborough, Hampshire]

T3 — Gist Summarization (EN). *Definition:* Produce a one-sentence English summary of the spoken content.

Input: “...if you don’t want to look at all of the man pages, you just want to see a very short description of what a command does—type whatis followed by the command name...”

Output: The whatis command provides a one-line description of any Unix/Linux command, e.g. whatis sort or whatis cat.

T4 — Topic Labeling. *Definition:* Assign a single short topic label (one to three words) to the utterance.

Input: “These data components in turn serve as the ‘building blocks’ of data exchanges.”

Output: data exchange

T5 — Numeric QA. *Definition:* Generate a question–answer pair where the answer is a number explicitly mentioned in the utterance.

Input: “Only one referendum has been held at the national level in Guyana.”

Output: Q: How many national-level referendums have occurred in Guyana? || A: 1 referendum

T6 — Gist Summarization (DE). *Definition:* Produce a one-sentence German gist summary of the spoken content.

Input: “...if you don’t want to look at all of the man pages, you just want to see a very short description of what a command does...”

Output: Der Befehl whatis liefert eine einzeilige Beschreibung eines Unix-Kommandos, etwa whatis ls oder whatis sort.

T7 — Gist Summarization (ZH). *Definition:* Produce a one-sentence Mandarin Chinese gist summary of the spoken content.

Input: “...if you don’t want to look at all of the man pages, you just want to see a very short description of what a command does...”

Output: 使用whatis命令可以快速查看Linux命令的简要描述。

T8 — Vocal Style Description (EN). *Definition:* Describe the speaker’s vocal style, tone, and delivery in English, inferred directly from the speech audio (no transcript used).

Input: [audio: “These data components in turn serve as the ‘building blocks’ of data exchanges.”]

Output: The speaker has a measured, confident tone, speaking at a moderate pace with clear articulation.

T9 — Vocal Style Description (DE). *Definition:* Describe the speaker’s vocal style in German, inferred directly from the speech audio.

Input: [audio: “These data components in turn serve as the ‘building blocks’ of data exchanges.”]

Output: Die Sprecherin klingt ruhig und professionell mit einem gleichmäßigen Sprechtempo und einer freundlichen, sachlichen Betonung.

T10 — Vocal Style Description (ZH). *Definition:* Describe the speaker’s vocal style in Mandarin Chinese, inferred directly from the speech audio.

Input: [audio: “These data components in turn serve as the ‘building blocks’ of data exchanges.”]

Output: 声音听起来平稳而中速，带有平静的语调，情感色彩比较客观。